

Pathogen

- disease-causing organism
- transmissible disease is a disease where the pathogen can be passed from one host to another
 - * can be transmitted...
 - by direct contact; through blood or ^{other} bodily fluids
 - indirectly; from contaminated surfaces, food, animals or air
 - the body defends itself with = skin, mucus, hairs in nose, stomach acid and WBC's
 - active immunity is the defense against pathogens by anti-body production
 - each pathogen has its own antigens, which have specific shapes
 - antibodies are proteins that binds to antigens leading to destruction of pathogens or marking of pathogens for phagocytes
 - specific antibodies have complementary shapes which fit specific antigens
- active immunity is gained after an infection by a pathogen or vaccination

process of vaccination:.

- weakened pathogens or their antigens are put in body
- the antigens stimulate an immune response by lymphocytes which produce antibodies
- memory cells are produced which give a long term immunity.

- vaccination helps in preventing the spread of disease by making you immune to it.

- passive immunity is a short term defense against pathogens by antibodies acquired from another individual, including across the placenta and in breast milk
- breast feeding is important for infants as it contains antibodies which help develop passive immunity.
- memory cells are NOT produced in passive immunity.

Cholera

- a disease caused by a bacterium which is transmitted in contaminated water
- the bacterium produces a toxin that causes secretion of chloride ions into the small intestine, causing osmotic movement of water, causing diarrhoea, dehydration and loss of ions from blood

Controlling the spread of disease:

- (1) clean water supply
- (2) hygienic food preparation
- (3) good personal hygiene
- (4) waste disposal
- (5) sewage treatment